A Major Project Report
(Phase-I)
On

AI-Driven Decision Support Avalanche Susceptibility Mapping System

Submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2026-27

in

Computer Science and Engineering

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May, 2026

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CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled **"AI-Driven Decision Support Avalanche Susceptibility Mapping System"**, in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University,** Greater Noida, is an original work carried out by us under the supervision of **Mr. Surinder Pal Singh**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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ABSTRACT

Avalanches present a serious and frequent danger to human life, to infrastructure, and to connectivity in high-altitude mountainous areas like the Indian Himalayas, since abrupt failures of snow slopes put at risk not only civilian populations but also tourists and defence personnel. Traditional avalanche forecasting depends greatly on manual field surveys, empirical snowpack models, and the judgement of experts, a method which is time-consuming, restricted in terms of spatial coverage, and hard to apply over large and remote areas. The present project introduces an AI-Driven Decision Support System for mapping avalanche susceptibility which combines satellite-derived remote sensing data, terrain parameters based on a Digital Elevation Model (DEM), and historical meteorological data within a machine learning and Geographic Information System (GIS) framework in order to produce spatially detailed avalanche susceptibility maps. The system uses a hybrid ensemble machine learning method that brings together algorithms such as Random Forest and XGBoost with GIS-based multi-criteria overlay analysis to divide the terrain into susceptibility zones varying from very low to very high risk. The main factors taken into account are slope angle, aspect, elevation, curvature, land cover, snow depth, and temperature variation. The decision support system thus produced is meant to help disaster management authorities, the Border Roads Organisation, and local governments in the Indian Himalayan region with the generation of early warnings, route planning, and the allocation of resources. This Phase-I report includes the problem definition, a literature survey, the objectives, the scope, and the feasibility study carried out to lay the groundwork for the proposed system, with an evaluation of the availability of data, the technical feasibility, and the multidisciplinary character of the project which spans the fields of computer science, geoinformatics, and disaster management.

1. Problem Definition

In the Indian Himalayan area, which includes provinces like Himachal Pradesh, Jammu & Kashmir, and Uttarakhand, snow avalanches occur frequently during the winter and early spring months. Such incidents often interfere with important highways such as the Srinagar-Leh and Manali-Leh routes, put military outposts at risk, and result in deaths among civilians, trekkers, and members of the armed forces. Yet, given the extent of the danger, avalanche hazard assessment in India still relies mainly on periodic field observations carried out by the Snow and Avalanche Study Establishment.

There is a need for a system that can provide high-resolution region-wide information about the chances of avalanches happening. The current methods for predicting avalanches have some problems. They are about reacting to what has already happened instead of predicting what might happen. They depend on information from points instead of data that covers a whole area. They also need a lot of knowledge to understand which means not everyone can use them easily. They don't bring together kinds of data like land shape, weather and what is on the land into one single report. They also can't grow big enough to cover the huge and hard to reach places in the Himalayas. So there is a need for a system that works on its own uses data in a smart way and can show the areas where avalanches are likely to happen. The results should be easy for people who make decisions to use. This project tries to fill that gap. It says the problem is this: given times when avalanches happened data about the land from a map that shows height and weather information for a place, in the Himalayas create a machine learning system that can sort the land into areas that are likely to have avalanches. Show these areas on a map that people can use easily. This helps with warnings and planning routes.

2. Literature Survey

A review of research and systems that deal with predicting avalanches and mapping where they are likely to happen was done. The goal was to find out what methods are being used what data is. Where there are missing pieces that this project will try to fix.

2.1 Statistical and Physical Snowpack Models

Traditional avalanche forecasting has relied on based snowpack simulation models such, as CROCUS and SNOWPACK. These snowpack simulation models simulate the structure of snow to estimate instability. While avalanche forecasting is physically grounded avalanche forecasting requires continuously updated field instrumentation and is computationally intensive. This makes avalanche forecasting impractical for generating susceptibility maps across data-sparse Himalayan terrain.

2.2 GIS-Based Multi-Criteria Decision Analysis

Many researchers have used GIS-based -criteria decision analysis (MCDA) to study land patterns. These researchers often mix terrain factors like slope, aspect, elevation and curvature together. They use weighted overlay techniques like the Analytic Hierarchy Process (AHP) to create susceptibility zonation maps. I think these GIS-based multi-criteria decision analysis (MCDA) methods are quite helpful because they are easy to see and understand. However there is a problem. These methods rely on weights that experts pick by hand. Because experts pick the

weights these GIS-based -criteria decision analysis (MCDA) methods might be biased. This bias can make the susceptibility zonation maps less accurate, than methods that use data.

2.3 Machine Learning Approaches

New studies show that machine learning classifiers like Random Forest, Support Vector Machines and Artificial Neural Networks work well for mapping where landslides and avalanches might happen. These researchers use things like the shape of the land and the weather to make these maps. Most of these studies find that group methods like Random Forest and XGBoost are better than using one single tool or old math models. Random Forest and XGBoost are better, at predicting where hazards will occur because Random Forest and XGBoost can understand the non-linear ways that land and weather interact.

2.4 Remote Sensing and Deep Learning

Researchers have used satellite pictures from Sentinel-1 SAR and Sentinel-2 optical sensors to show how smart computer programs can find avalanche debris and snow cover changes on their own. This helps because people do not have to go out and check everything by hand.. There is a big problem with these smart computer programs. They usually need an amount of labeled data to work well. Now there is not enough labeled data, for the Indian Himalayan area.

2.5 Research Gap

In the studies that I have read many researchers note that there is no complete decision support system specifically designed for the Indian Himalayan region. This missing tool would bring together terrain analysis, weather information and machine learning inside one easy-to-use GIS screen that people who are not experts—, like disaster managers—could operate. My project aims to fill that void by building a ML-GIS decision support system.

3. Objectives

The primary aim of this project is to design and develop an AI-driven decision support system for avalanche susceptibility mapping in the Indian Himalayan region. The specific objectives are as follows:

- To find and collect the land shape, weather and past avalanche data that we need to make avalanche models, in the chosen study region.
- To extract and process terrain-derived conditioning factors (slope, aspect, elevation, curvature, and land cover) from Digital Elevation Model and satellite data using GIS tools.
- To design and train a hybrid machine learning model (Random Forest / XGBoost ensemble) capable of classifying terrain units into discrete avalanche susceptibility classes.
- To validate the trained model against historical avalanche occurrence records and evaluate performance using standard metrics such as accuracy, precision, recall, and the Area Under the ROC Curve (AUC-ROC).
- To develop an interactive GIS-based decision support interface that visualises the generated susceptibility map and presents actionable outputs for disaster management stakeholders.
- To assess the feasibility, scope, and multidisciplinary requirements of the system as part of the Phase-I project initiation study.

4. Scope

The project covers data engineering, building machine learning models and creating maps using GIS. It is planned in a way that it can be done during the time of a B.Tech project. It is also designed to allow work in the future.

4.1 In Scope

- Collection and preprocessing of publicly available DEM (e.g., SRTM/ASTER), satellite imagery (Sentinel-1/2), and meteorological datasets for a selected Himalayan study area.
- Extraction of terrain conditioning factors and construction of a structured feature dataset suitable for supervised machine learning.
- Development, training, and evaluation of ensemble machine learning models for susceptibility classification.
- Design of a GIS-based map visualisation layer to display susceptibility zones with an intuitive colour-coded interface.
- Basic decision-support outputs such as risk-zone summaries to aid route planning and early alerts.

4.2 Out of Scope

- Real-time sensor network deployment and live IoT-based snowpack monitoring, which is proposed as future work beyond Phase-I/II.
- Nationwide coverage; the initial system will be developed and validated for one representative Himalayan study region.
- Integration with government emergency response communication systems, which would require institutional partnerships beyond the academic project scope.

4.3 Novelty and Multidisciplinary Aspects

The new idea of the project is to bring many different machine learning methods with GIS-based ways of looking at the land and different factors that affect the terrain. This is done in one system that helps people make decisions. This system is made for the Indian Himalayan area. There is not a tool like this that is easy for people who manage disasters in different regions. The project needs different areas of knowledge. It uses computer science and machine learning. It uses geoinformatics. Taking pictures from the sky. It also uses knowledge about disasters and dangers. The team has to work with data science methods. They use tools for looking at maps and the land, like QGIS and ArcGIS. They also use Python programs for working with map data. They also need to understand how snow slides happen.

5.5 Product Development Scope and Bill of Materials

As a primarily software-based decision support system, the project does not require physical hardware procurement. The core components and Bill of Materials (BoM) for development are summarised below.

Component	Description / Source
Digital Elevation Model (DEM)	SRTM 30m / ASTER GDEM (open-source)
Satellite Imagery	Sentinel-1 (SAR) & Sentinel-2 (optical), Copernicus Open Access Hub
Meteorological Data	IMD / open climate reanalysis datasets (e.g., ERA5)
Historical Avalanche	SASE reports and published research datasets

Records	
GIS Software	QGIS (open-source), GDAL, Python geospatial libraries
ML/Development Environment	Python, scikit-learn, XGBoost, Jupyter/Google Colab
Visualization/Decision Support Interface	Web-based GIS map viewer (e.g., Leaflet/Folium)

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